

Wearable Instrument:

An Intimate Dialogue Between Body, Sound, and Garment

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Figure 1: Final shot, photo credit: Mina Anttila



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Abstract

This paper explores the transformation of knitwear into wearable musical instruments through the embedding of stretch-responsive knitted sensors in garments. The aim is to reconnect the body and sound through textiles, particularly for non-experts in music, in the context of the digitisation of music and the lack of physical, tactile touch and interaction. It addresses a gap in the field of smart wearables, where functionality often takes precedence over aesthetics and emotional experience.

The final product is a knitted dress that integrates seven knitted resistive sensors, which are connected to a sound synthesis system built using Pure Data (Pd). Each sensor is mapped to specific sound parameters, such as pitch, reverb or frequency, which are triggered by natural body movements.

Ccs Concepts

•Applied computing~Arts and humanities~Sound and music computing~Performing arts •Human-centered computing~Interaction design~Interaction design process and methods~Participatory design~User centered design •Hardware~Communication hardware, interfaces and storage~Wireless integrated network sensors~Sensor devices and platforms~Sound-based input / output

Keywords

smart wearables; wearable instrument; e-textiles; stretch sensors; embodied interaction; sound design; knitting; fashion technology

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1 Introduction

As digital technologies and MIDI advancements continue to evolve, our interaction with sound has become increasingly mediated by screens and devices, often at the expense of tactile engagement. This shift raises critical questions about the sensory disconnection in modern sound interaction and the need for more intuitive and embodied approaches to musical expression. In response to this challenge, this project investigates the potential of wearable sound technologies to reconnect users with the tactile and performative aspects of music creation. By

integrating knitted, stretch-responsive sensors into a knitted garment, this project seeks to bridge the gap between sound design and e-textile innovation, with a focus on accessibility, wearability, and democratization of musical expression, enabling non-experts to create sound through natural and intuitive body flow. [9, 18]

2 Design Concept & Motivation

My background in fashion design has shaped my perspective on how garments interact with the body, tactically, ergonomically, and aesthetically. This project started from a desire to extend fashion beyond style and function, transforming it into an interface for musical and performative expression.

While much of the existing work in smart wearables is driven by engineering goals, many devices lack sensitivity to the cultural, sensory, and aesthetic contexts of the body. Wearables such as Waisvisz’s *The Hands* or Maubrey’s *Audio Peacock* [1,2] explore sound-based interaction, but often feature bulky or exposed components.[15, 12] Inspired by this tradition yet seeking a softer, more intimate approach, I turned to knitwear as a design medium.

Knitting provides natural elasticity, softness, and structural flexibility due to its loop-based construction, qualities not typically found in woven fabrics. This makes knitted textiles especially suitable for embedded stretch sensors and garments meant for expressive movement[10, 18]. The softness of the knitted garment encourages the wearer to move intuitively, embracing embodied sound-making without technical mediation.

I also drew from feminist HCI principles as outlined by Bardzell [2], which emphasize affect, inclusion, and lived experience in interaction design. These principles encouraged me to design for emotional resonance, accessibility, and playfulness—especially for non-musicians. Moreover, embodied ideation, as advocated by Wilde and Tomico [20], played a key role in my prototyping. By moving and listening in the garment, I designed iteratively with my body as part of the sound system.

3 Implementation Details

3.1 Sensor Mapping and Sound Logic

- **Armpit and elbow sensors:** Each mapped to 5 note triggers based on the readout range, enabling one-octave performance in C minor scale.
- **Skirt Sensors:** When stretched, they act as frequency filters for background soundtracks—higher stretch lets through only low frequencies, functioning like a volume controller.
- **Back Sensor:** Hunching triggers a reverb effect.
- **Metal Closure:** Main on/off switch that fades in/out the whole track.

3.2 Sound Synthesis

I used Pure Data for granular synthesis, mapping amplitude, pitch, and timbre to motion intensity. The signal path ensures that multiple sensors can act independently but harmoniously, preserving musical clarity even during layered input.

3.3 Yarn & Knit

In knitting, every choice and detail impacts the final results. The main goals for this project are to create a fabric that is thin, sleek, and soft, while also being flexible and able to recover its shape for stable sensor performance after repeated use. To achieve thinness and sleekness, background structures like half Milano, interlock, and single jersey were tested. For flexibility and recovery, elastic yarns and structures like ribs and links links were experimented.

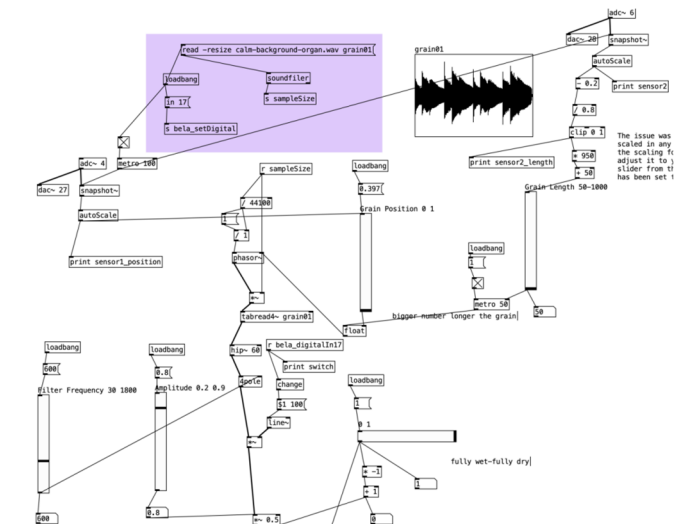


Figure 2: Sound programming in puredata

Different directions of stretchiness were created in knitting structures due to the need on different body parts. For example, to make a knitted sleeve with stretching sensor at elbow, the sensor part should be stretchy vertically and the background should be stretchy horizontally. In this case, I tried out different knitting structures, stitch lengths, and yarns to make sure the one perform the best for the purpose.

Intasia as a knitting structure allows an embedded pattern with different yarns and structures, which is perfect in this case. As shown in the pictures, I tried some variations of “links-links” structure, including 1×1 (one row repeat for front and rear bed), 2×2, 3×3; 3×3 check, 9×9 check; pointelle, etc.; and some structures for the background, such as ribs, half milano, alternative single jersey, interlock, RR, etc.

Besides various structures, different colorways and sensor shapes were tested as well. The final color combination and structures were determined after extensive sampling. Brown and red were chosen for contrast, with 2×2 ribs used for the main body, half milano for the bag and 2×2 links-links for the sensor areas.

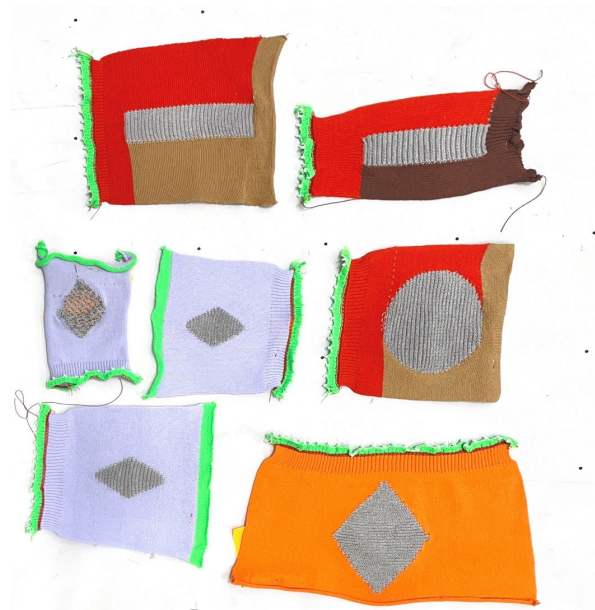


Figure 3: Knitting samples and sensor prototypes. Links-links and rib structures tested for consistent stretch and return behavior. Conductive yarns embedded using intarsia knitting.

From top to bottom, left to right: 2*2 links links (sensor) and half milano (background); 3*3 links links (sensor) and 2*1 ribs (background); Pointelle (sensor) and single-bed (background); the 4th to 7th samples: 2*2 links links (sensor) and half milano (background), with different color combinations, stitch length and sensor shapes.

3.4 Sketch & Form

I created sketch overlays to map movement zones to sound types (Figure 4). The final shape ensures comfort, freedom of movement, and aesthetic balance. The sound logic guides the choreography of interaction.

3.5 Integration of Electronics into Design

The electronics also function as decorative design elements. To enhance the fashion-oriented appearance and improve wearer comfort, I chose to conceal the wires within ribbons made from the same yarns. A back-mounted bag will house the processors (Raspberry Pi or Bela board), with the circuit board placed inside a patch pocket. This design choice minimizes visual exposure of electronic components, shifting the aesthetic away from an engineering-heavy look toward a more integrated and wearable design.

Each sensor requires two wires to detect the stretching value before and after movement. To streamline the design, each pair of wires is combined into a single tunnel, resulting in eight decorative tunnels that are hand-sewn onto the garment.

To enhance silhouette smoothness and comfort, the back-mounted bag includes padding. The wire cover tunnels feature

the same filled construction, the top and skirt are also connected with these structured tunnels.



Figure 4: Conceptual sketch showing body-sensor mapping.

4 Discussion

This project proposes an alternative to screen-based interfaces by situating sound-making within embodied, tactile gestures. The wearability and design-forward construction offer new accessibility in digital music-making, especially for non-experts. Limitations include potential instability in sensor calibration, and fit-specific tuning across different body sizes. Future work includes exploring modular sensor systems and integrating washable, flexible PCB materials.

4.1 Fit

Since the sensor placements must be highly precise for each body part, different wearers might experience fit issues, even though the stretchy knit material allows for a broad range of body sizes. Ensuring accurate sensor alignment remains a challenge.

A potential solution could be modular sensor placement, allowing wearers to adjust and assemble the sensors onto the garment themselves. In this case, using woven fabric with a cut-and-sew construction might be a more suitable alternative, providing better adaptability for different body shapes.

4.2 Stability

With continuous body movements, the risk of connection instability increases. A key challenge is securing wire connections while maintaining freedom of movement for the performer. Further research is needed to develop durable yet

flexible wiring solutions that can withstand dynamic performances without compromising functionality.

Another risk to stability is the sensor's elasticity. If the sensor is stretched and doesn't return to its original state quickly enough, the value range may change, leading to inaccuracies in controlling the synthesis.

4.3 Customized PCB

A customized PCB (Printed Circuit Board) not only enhances functional performance but also contributes to the aesthetic integration of the circuitry into the garment. Instead of using a traditional PCB, alternative approaches include:

- Printed PCBs using conductive materials
- Custom PCB design using software like KiCad
- PCB milling techniques to carve the circuit

Exploring custom PCB designs could offer greater creative possibilities, ensuring the electronic components blend seamlessly into the garment's overall design language.

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